

DES Key Generation Using Two 28-Bit Halves

18. Write a C program for DES the first 24 bits of each subkey come from the same subset of 28 bits of the initial key and that the second 24 bits of each subkey come from a disjoint subset of 28 bits of the initial key.

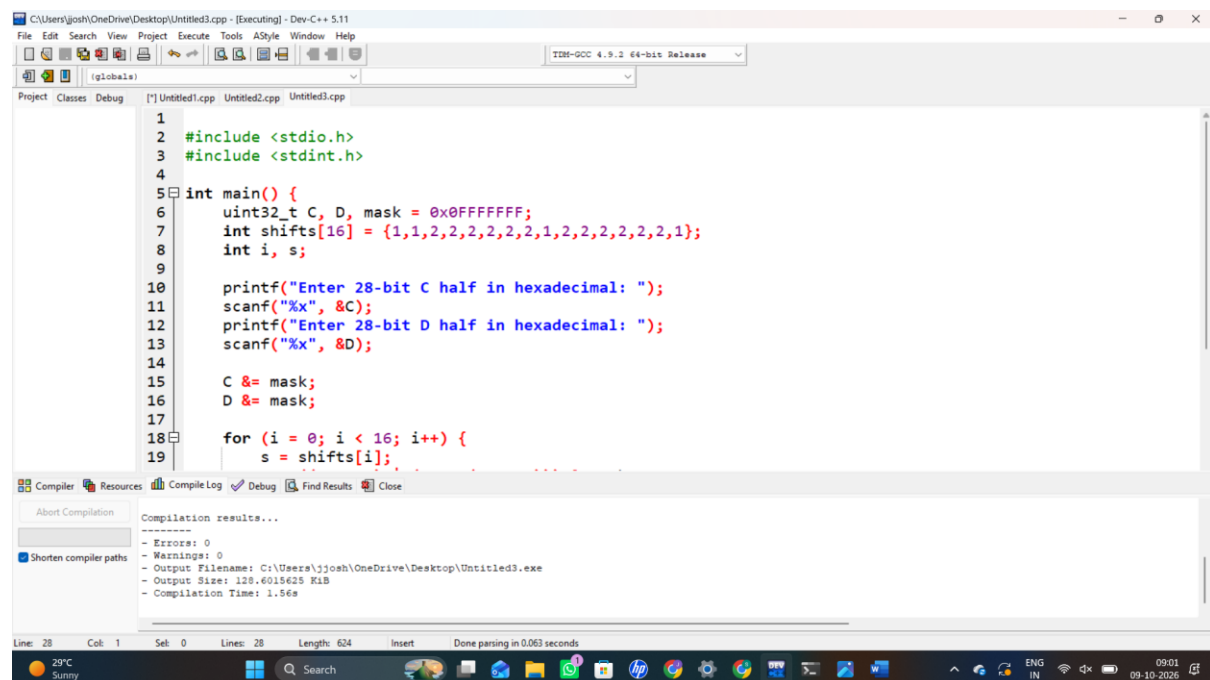
Aim:

To write a C program demonstrating DES key generation by dividing the 56-bit effective key into two 28-bit halves and rotating them according to the DES shift schedule.

Algorithm:

1. Start the program.
2. Read two 28-bit halves, C and D.
3. Apply the DES left-shift schedule for 16 rounds.
4. Display the rotated halves for each round.
5. Stop.

Program :



```
1
2 #include <stdio.h>
3 #include <stdint.h>
4
5 int main() {
6     uint32_t C, D, mask = 0xFFFFFFFF;
7     int shifts[16] = {1,1,2,2,2,2,2,2,1,2,2,2,2,2,1};
8     int i, s;
9
10    printf("Enter 28-bit C half in hexadecimal: ");
11    scanf("%x", &C);
12    printf("Enter 28-bit D half in hexadecimal: ");
13    scanf("%x", &D);
14
15    C &= mask;
16    D &= mask;
17
18    for (i = 0; i < 16; i++) {
19        s = shifts[i];
```

Compilation results...

Errors: 0
Warnings: 0
Output Filename: C:\Users\jjoah\OneDrive\Desktop\Untitled3.exe
Output Size: 120.6015625 KiB
Compilation Time: 1.56s

Input/Output:

```
C:\Users\josh\OneDrive\Desktop >
Enter 28-bit C half in hexadecimal: 1234567
Enter 28-bit D half in hexadecimal: 7654321
Round 1: C = 2468ACE, D = ECA8642
Round 2: C = 48D159C, D = D958C85
Round 3: C = 2345671, D = 6543217
Round 4: C = 8D159C4, D = 958C85D
Round 5: C = 3456712, D = 5432176
Round 6: C = D159C48, D = 58C85D9
Round 7: C = 4567123, D = 4321765
Round 8: C = 159C48D, D = 0C85D95
Round 9: C = 2B3891A, D = 198BB2A
Round 10: C = ACE2468, D = 642ECA8
Round 11: C = B3891A2, D = 98BB2A1
Round 12: C = CE2468A, D = 42ECA86
Round 13: C = 3891A2B, D = 0BB2A19
Round 14: C = E2468AC, D = 2ECA864
Round 15: C = 891A2B3, D = BB2A198
Round 16: C = 1234567, D = 7654321

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Process exited after 20.8 seconds with return value 0
Press any key to continue . . .
```

Result:

The program demonstrates the DES 16-round key-schedule rotations on two 28-bit halves.